Padraig Lavery

Forward Deployed Engineer – Agentic AI, Cloud & Production Systems

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years building, deploying and running production systems for UK and Middle East banking clients on AWS and Kubernetes, across separate deployments for each bank. Building applied LLM skills hands-on: Incident Copilot is an agentic, retrieval-based triage service being built on locally hosted models. Brings the production habits agentic platforms need in client environments: safe failure, clear runbooks, security and compliance. Preparing for the AWS Certified AI Practitioner exam (November 2026).

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Deploying into client environments

- Planned testing for the AWS SDK v3 and Node 22 Lambda runtime upgrades across repositories and multiple bank deployments, and authored the engineering migration guide for Lambdas, Fargate services and microservices.
- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals to meet each client's security and compliance audit requirements.
- Led the technical migration of email templates onto dedicated brand domains across four banking brands, preserving deliverability.
- Resolved a live customer-facing outage in a Kubernetes/Istio ingress setup and permanently corrected the Helm deployment templates.

Building systems that survive production

- Redesigned a month-end invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation, so one failing partner no longer aborts the run.
- Eliminated recurring daily out-of-memory crashes in a high-volume ingestion service by loading caches once per batch instead of per file.
- Hardened authentication and account services and remediated critical Snyk-flagged vulnerabilities for PCI DSS compliance.
- Authored runbooks and SOPs so the production services team resolves recurring issues without engineering escalation.

AI and delivery

- Built the CI/CD pipeline for a UI Builder service as part of an internal AI investigation project.
- Led a cross-team initiative for automated API testing (validation, shared error handling, OpenAPI docs, unit tests, stable GitHub Actions CI).
- Mentored four new starters; run the team's retrospectives and author technical guides.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a cloud-cost web API using Go, Docker, Kubernetes and Git.

AI PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Incident-triage service: takes a production incident description, retrieves relevant runbook content and suggests likely root causes and next steps, using agentic tool-calling to check live service status.
- Moved inference from AWS Bedrock to locally hosted models via Ollama to remove inference costs. Runbook knowledge base and FastAPI service in place; retrieval, model integration and testing in development.

AI-Based Credit Card Fraud Detection (final year project) | Angular, Flask, GraphQL, Docker, Kubernetes, AWS SageMaker

- End-to-end web app flagging irregular card transactions, with a model trained in R and Python and served via SageMaker.

SKILLS

- **AI:** LLM integration, retrieval-augmented generation, agentic tool-calling, pgvector, Ollama, AWS Bedrock, SageMaker
- **Languages:** Python, JavaScript (Node.js), C#, Go, SQL

- **Back end:** FastAPI, Node.js, .NET Core, microservices, OpenAPI
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **Practices:** production incident response, runbooks, security remediation (Snyk, OWASP)

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026